

VIDEO-LLM FOR PROACTIVE SAFETY MONITORING

Keynote 3 — Afshin Dehghan (Apple)
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Problem

A safety system should understand what is happening while a video is being watched — not only after the recording is finished.

Inspired by the keynote's focus on streaming and proactive Video-LLM assistants.

The idea

Use a Video-LLM as the reasoning layer for a simple safety monitor.

- 1 Watch**
Sample a few frames from the video stream.
- 2 Understand**
Ask the Video-LLM what is happening in the clip.
- 3 Respond**
Trigger an alert only when an important event is detected.

Why this matches the keynote

The keynote describes a move from training-free video understanding toward streaming Video-LLMs that can reason over longer video and support proactive responses.

Prototype goal

Can a small Video-LLM watch a short video, describe the activity, and decide whether there is a safety-related event?

Quick prototype: it worked

Real local run using LLaVA-OneVision 0.5B on CPU.

INPUT

9.72-second video • 640 × 360 • 243 frames • 8 sampled frames

ACTUAL MODEL OUTPUT

“Reading a book”

Safety check: “No important or safety-related event happened.”

Feasibility result: the basic Video-LLM workflow works end-to-end.

No accuracy score is reported because this quick test did not use a labeled ground-truth set.

What this shows

The model successfully processed a video and produced both a description of the activity and a simple safety judgment.

What comes next

Test longer videos, faster models, streaming input, and real alert rules.